



Genome-wide analysis of the *Hsf* gene family in *Brassica oleracea* and a comparative analysis of the *Hsf* gene family in *B. oleracea*, *B. rapa* and *B. napus*

Neeta Lohani¹ · Agnieszka A. Golicz¹ · Mohan B. Singh¹ · Prem L. Bhalla¹ 

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Abstract

The global climate change-induced abiotic and biotic stresses are predicted to affect crop-growing seasons and crop yield. Heat stress transcription factors (Hsfs) have been suggested to play a significant role in various stress responses. They are an integral part of the signal transduction pathways that operate in response to environmental stresses. *Brassica oleracea* is one of the agronomical important crop species which consists of cabbage, cauliflower, broccoli, Brussels sprout, kohlrabi and kale. The identification and roles of Hsfs in this important *Brassica* species are unknown. The availability of whole genome sequence of *B. oleracea* provides us an opportunity for performing in silico analysis of *Hsf* genes in *B. oleracea*. Thirty-five putative genes encoding Hsf proteins were identified and classified into A, B and C classes. Their evolution, physical location, gene structure, domain structure and tissue-specific expression patterns were investigated. Further, a comparative analysis of the *Hsf* gene family in *B. oleracea*, *B. rapa* and *B. napus* highlighted the role of hybridisation and allopolyploidy in the evolution of the largest known *Hsf* gene family in *B. napus*. The presence of orthologous gene clusters, found in *Brassica* species, but not in *A. thaliana*, suggested that polyploidisation has resulted in the formation of new *Brassica*-specific orthologous gene clusters. Gene duplication analysis indicated that the evolution of the *Hsf* gene family was under strong purifying selection in these *Brassica* species. High-level synteny was observed within the *B. napus* genome. Conservation of physical location, the similarity of structure and similar expression profiles between the *B. napus* *Hsf* genes and the corresponding genes from *B. oleracea* and *B. rapa* suggest a high functional similarity between these genes. This study paves the way for further investigation of *Hsf* genes in improving stress tolerance in *B. oleracea*. The genes thus identified may be useful for developing crop varieties resilient to the global climate change.

Keywords *Hsf* genes · *Brassica* · Polyploidy · Abiotic stress · Heat stress

Introduction

Plant development is affected by abiotic stresses such as heat, cold, drought and salinity. Plants being sessile have evolved adaptation strategies to cope with stresses. Timely and

efficient response to abiotic stress is a key to plant survival, crop productivity and yield loss. Transcription factors (TFs) such as Heat Shock Transcription Factors (Hsfs), Dehydration-Responsive Element Binding Protein 2A (DREB2A), Multipurpose Binding Factor 1C (MBF1C) and NAC019 are involved in the transcriptional regulation of stress response in plants (Ahuja et al. 2010; Ohama et al. 2017; Singh et al. 2002). These transcription factors regulate the expression of chaperones such as heat shock proteins (HSPs), reactive oxygen species (ROS), scavenging enzymes (ascorbate peroxidase, APX, catalase, CAT) and several other genes involved in signal transduction and regulatory pathways (Ohama et al. 2017). Moreover, Hsfs have been shown to play an important role in heat stress response and other abiotic and biotic stresses. Hsfs regulate the expression of stress-related

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✉ Prem L. Bhalla
premlb@unimelb.edu.au

¹ Plant Molecular Biology and Biotechnology Laboratory, Faculty of Veterinary and Agricultural Sciences, University of Melbourne, Parkville, Melbourne, VIC, Australia

genes and chaperone proteins such as HSPs. HSPs play a critical role in restoration of cellular homeostasis and defence mechanism in plants in response to various environmental stresses (Guo et al. 2014). Various HSP families such as HSP17, HSP70 and HSP90 are upregulated by HSFs under stress conditions in crop plants (Jung et al. 2013; Kumar et al. 2018). Further, *Hsf* genes are also involved in plant development (Almoguera et al. 2002; Díaz-Martín et al. 2005; Kotak et al. 2007). Understanding of the molecular basis of abiotic stress tolerance, including the function of Hsfs, is especially important now as the global climate change is predicted to affect the crop-growing seasons and hence crop yield. A detailed account of *Hsf* genes in the economically important crops is required for developing crop improvement strategies.

Studies on various organisms have indicated that Hsf proteins are composed of structurally and functionally conserved domains—DNA binding domain (DBD) at the N-terminal, an adjacent bipartite oligomerization domain (HR-A/B or OD), nuclear localization signal (NLS), nuclear export signal (NES), C-terminal activator peptide protein (AHA) and repressor domain (RD) (Guo et al. 2016). Of these conserved domains, DBD has been the most preserved domain among Hsfs. DBD is characterised by a central helix-turn-helix motif, and it is responsible for binding to the heat shock elements (HSEs) located in the promoters of the target genes (Scharf et al. 2012). The HSEs are palindromic binding motifs conserved in promoters of heat stress-inducible genes (HSEs: 50-AGAAnnTTCT-30) (Scharf et al. 2012; von Koskull-Döring et al. 2007). On the other hand, the OD domain is composed of hydrophobic heptad repeats and is present in all the Hsfs (Nover et al. 2001; Scharf et al. 2012).

The *Hsf* gene family can be divided into three classes: A, B and C (Guo et al. 2016). Along with the flexible linkers, the OD domain also forms a basis of classification. Class A and C Hsfs contain 21 and seven additional amino acid residues in the HR-A/B region, respectively (Schmidt et al. 2012; Wang et al. 2014), which are absent in class B Hsfs.

Hsfs only in their active state are capable of activating target genes. The inactive monomer state of Hsfs is changed to an active state by oligomerisation, which is initiated by stress conditions (Hartl and Hayer-Hartl 2002). The repeats present in the OD allow homo- or hetero-oligomerization of the Hsfs (Baniwal et al. 2004).

The functional activity of Hsfs is governed by the shuttling balance between the cytoplasm and nucleus (Kotak et al. 2004). The nuclear localization signal (NLS) and the nuclear export signal (NES) mediate the intracellular distribution and interactions of Hsfs between the nucleus and cytoplasm. AHA motifs are specific to the HsfA subfamily and confer transcriptional activator function. HsfB subfamily members act as repressors (Czarnecka-Verner et al. 2004) and are characterised by the presence of a repressor domain (RD).

In comparison to animals and fungi, plants contain a number of Hsfs. Yeast and *Drosophila* have been reported to carry one *Hsf* gene (Sorger and Pelham 1988) and *Homo sapiens*, three *Hsf* genes (Pirkkala et al. 2001). To date, the *Hsf* gene family has been identified in several agronomically important plants such as rice, wheat, maize, soybean, legumes and cotton (Chung et al. 2013; Lin et al. 2011; Lin et al. 2014; Mittal et al. 2009; Wang et al. 2014; Xue et al. 2013). Fifty-six *Hsf* genes were reported in wheat (Xue et al. 2013), and the largest *Hsf* gene family has been identified in *B. napus* with 64 *Hsf* genes (Zhu et al. 2017).

The allopolyploid *B. napus* was formed as a result of the natural hybridisation of genomes of *B. rapa* and *B. oleracea* approximately 7500 years ago (Chalhoub et al. 2014). The more ancient polyploidisation events (Adams and Wendel 2005; Cui et al. 2006), along with the recent hybridisation, shaped the *B. napus* genome and resulted in high gene content of ~100,000 genes, distributed on 19 chromosomes (9 chromosomes belonging to the C_n subgenome and 10 to the A_n subgenome). Polyploidisation, hybridisation and the subsequent gene loss govern the size of the entire gene complement as well as the individual gene families (De Grassi et al. 2008). In *Arabidopsis*, certain transcription factor families have been shown to be influenced by segmental duplications (Cannon et al. 2004). In *B. oleracea*, the whole genome triplications (WGT) resulted in over-retention of transcription factors (Liu et al. 2014).

Hsf genes constitute an important family of transcription factors which play critical roles and act as central regulators in plant responses to different stresses. Therefore, it would be interesting to examine how the hybridisation and allopolyploidy have affected the evolution of the *Hsf* gene family in *B. napus*. As a prerequisite, the characterisation and identification of the *Hsf* gene family members in all the three species (*B. napus*, *B. rapa* and *B. oleracea*) is required. The *Hsf* gene family has been comprehensively described and studied in *B. rapa* (Song et al. 2014) and *B. napus* (Zhu et al. 2017), but not in *B. oleracea*.

In this study, we aim first to carry out an in silico identification of the *Hsf* gene family in *B. oleracea* and then to study the structural and evolutionary relationship between the *B. oleracea*, *B. rapa* and *B. napus* *Hsf* gene family. A total of 35 putative genes were identified and characterised as members of the *Hsf* gene family from *B. oleracea* followed by a comparative investigation to study the evolution of the *Hsf* gene family in *B. napus*. Gene duplication analysis indicated the extent to which duplication events resulted in the expansion of the *Hsf* gene family. The synteny between the genomes of *B. napus* and *B. oleracea* and *B. napus* and *B. rapa* was also studied. Then, those syntenic genes were analysed for structural similarities, and the expression level of these genes in various tissues was compared. This study highlights the functional and evolutionary history of *Hsf* genes

in *Brassicaceae* and enhances our understanding of the effect of hybridisation and polyploidisation events on the evolution of the *Hsf* gene family.

Materials and methods

Identification and characterisation of HSFs in *B. oleracea*

The protein and nucleotide sequences of *B. oleracea* were downloaded from EnsemblPlants (*Brassica_oleracea.v2.1.36*). The conserved domain of Hsf is Hsf-Type DBD domain. The Hsf-type DBD domain (Pfam: PF00447) was submitted as a query in a BLASTP search of the *B. oleracea* proteome. TBLASTN searches were performed using the putative Hsf protein sequences as query against the NCBI nucleic-acid sequence data repositories in order to identify transcript isoforms (different version of transcripts representing the same gene). The Plant Transcription Factor Database 4.0 (Jin et al. 2016) (<http://plantfdb.cbi.pku.edu.cn/>) database was also searched using the search term ‘HSF’ and HSF domain search model accession ‘PF00447.12’. The amino acid sequences of Hsfs were extracted. The SMART 7 software (Letunic et al. 2011) (<http://smart.embl-heidelberg.de/>) was used to identify integrated DBD domain and (HR-A/B) domain in the putative Hsfs. Candidate proteins without integrated DBD domain and HR-A/B domain were removed. Physical properties of the predicted Hsf proteins such as protein isoelectric point (pI), molecular weight (Mw), instability index, aliphatic index and grand average of hydropathicity (GRAVY) were also analysed using the ProtParam tool of ExPASy server (Gasteiger et al. 2005) (<https://web.expasy.org/protparam/>). Chromosomal location, genomic, coding and peptide sequences were downloaded from the EnsemblPlants database. A distinctive name for each of the *Hsfs* identified in *B. oleracea* was given according to the classification of *Hsfs* in classes A, B and C. The *Hsf* genes in *B. oleracea* were referred to as *BoHsf* genes.

Multiple sequence alignment and phylogenetic analysis

Multiple sequence alignment was carried out to identify the highly conserved regions of the *BoHsf* proteins. ClustalX 2.0 (Larkin et al. 2007) and MUSCLE (Edgar 2004) software were used to align the *BoHsf* protein sequences to identify DBD and OD domains. A total of 35 *B. oleracea* HSFs, 21 *A. thaliana* HSFs (Scharf et al. 2012), 36 *B. rapa* Hsfs (Song et al. 2014) and 64 *B. napus* Hsfs (Zhu et al. 2017) were included in the phylogenetic analysis. The phylogenetic tree

was constructed by the MEGA 7.0 software using the neighbour-joining option and pairwise deletion. Bootstrap analysis was performed with 1000 replicates to assess the level of statistical support for each tree node (Kumar et al. 2016). The evolutionary distances were computed using the Poisson correction method (Kumar et al. 2016) and are in the units of the number of amino acid substitutions per site.

Gene structure and domain analysis

The domain analysis was performed to analyse the most important conserved, i.e. DBD, OD, NLS, NES and AHA, domains using different software tools. NLS domains in *BoHsfs* were predicted using the cNLS Mapper software (Kosugi et al. 2009) (http://nls-mapper.iab.keio.ac.jp/cgi-bin/NLS_Mapper_form.cgi/). NES domains were predicted using the NetNES 1.1 server software (La Cour et al. 2004) (<http://www.cbs.dtu.dk/services/NetNES/>). AHA domains were predicted based on the conserved-type AHA motif sequence FWxxF/L, F/I/L (Kotak et al. 2004). To further refine the search for conserved domains, the Heatster database (Scharf et al. 2012) (<http://www.cibiv.at/services/hsf/>) was used to analyse all the functional domains and to further classify the members of the *B. oleracea* Hsf family based on structural characteristics.

Sequence-based identification of orthologous *Hsf* genes

The orthologous genes in *B. oleracea*, *B. rapa* and *B. napus* were identified using OrthVenn (Wang et al. 2015) (<http://www.bioinformatics.net/OrthoVenn>). The protein sequences of the identified *Hsf* genes from all the three species were used for the analysis. Similar analysis was also done after including *A. thaliana* Hsf protein sequences. Also, each *Brassica* species was individually analysed with *Arabidopsis* sequences for identification of orthologous gene clusters.

Gene duplication analysis

Gene duplication was investigated in all three *Brassica* species (*B. oleracea*, *B. rapa* and *B. napus*) using Jalview (Waterhouse et al. 2009). Evolutionary distances between the genes were calculated by ClustalX 2.0. *Hsf* gene duplication was indicated by two criteria: (a) shared aligned nucleotide sequence covering > 80% of the longer gene and (b) similarity of the aligned regions of protein $\geq 80\%$ (Wang et al. 2014). Ancient duplication events were also studied to confirm the gene duplication events. The *Ka* and *Ks* values were calculated. The *Ka* and *Ks* were computed between pairs of genes identified as homeologous by MCScanX. Proteins were aligned by Clustal Omega 1.2.0 (Sievers et al. 2011). The protein alignments were converted to nucleotide alignments

using PAL2NAL v14 (Suyama et al. 2006). The Ks values were calculated using PAML (yn00) v4.7 (Yang 2007).

Synteny analysis

Genome annotations and corresponding protein sequences were downloaded from EnsemblPlants (Brassica_rapa.IVCAASv1.36, Brassica_oleracea.v2.1.36, Arabidopsis_thaliana.TAIR10) and GenoScope (Brassica_napus.annotation_v5). Synteny detection was performed using MCScanX (Wang et al. 2012) using default parameters. All-vs-all protein sequence comparisons necessary for MCScanX were performed using DIAMOND v0.8.25 (Buchfink et al. 2015) (–max-target-seqs 5 –evalue 0.00001).

RNA-Seq expression analysis

The RNA-Seq data was downloaded from the NCBI Sequence Read Archive (SRA, *B. napus*: ERR431515, ERR431516, ERR431517, ERR431518, ERR431522, ERR431524, ERR431525, SRR3571664, SRR3571674; *B. oleracea*: SRR630923, SRR630924, SRR630925, SRR630927, SRR630922, SRR630926, SRR630928; *B. rapa*: SRR643622, SRR643623, SRR643624, SRR643625, SRR643626, SRR643628, Supplementary Table 1). The sequencing data used for expression analysis was based on tissues collected from *B. rapa*, *B. oleracea* and *B. napus* plants grown in non-stressed conditions. These RNA-Seq data sets were used for tissue-specific expression analysis of the *Hsf* genes in the three *Brassica* species. The reads were mapped to the corresponding genomes (Brassica_rapa.IVCAASv1.36, Brassica_oleracea.v2.1.36, Brassica_napus_v4.1) using Hisat2 v2.0.5 (Kim et al. 2015). Reads mapping to HSF genes were counted using featureCounts v1.5.1 (Liao et al. 2013). Read abundance was calculated as transcripts per kilobase million (TPM).

Results

Identification of HSFs in *B. oleracea*

The amino acid sequences of the putative Hsfs were extracted from the *B. oleracea* (T1000) genome database using the conserved Hsf domain (PF00447). Also, the PlantTFDB and searches using the BLASTP program resulted in identification of putative *Hsf* gene candidates. In total, 67 amino acid sequences were retrieved as putative Hsfs. The transcript isoforms were removed based on the TBLASTN search results. A total of 35 putative *BoHsf* gene candidates were identified after detailed analysis of the conservation of the DBD domain and the presence of the coiled-coil structure using the SMART database.

The *BoHsf* genes were distributed on all the chromosomes (C01 to C09); however, the number of *Hsf* genes on every chromosome varied (Table 1). The physical location of one gene (*BoHsf04*) was not known and hence it was found on an unplaced scaffold. Chromosome 3 is the largest chromosome in *B. oleracea* both in terms of length of the chromosome (in bps) and the number of genes. Chromosome 3 had the largest number of *Hsf* genes, i.e. eight, followed by C01, C09 and C07 with five *Hsf* genes each. The lowest number was found on chromosomes 2, 4, 6 and 8 (two *Hsf* genes each). One of the *BoHsf* genes, *BoHsf30*, did not have a gene id in the *B. oleracea*.v2.1.36 (TO1000) genome annotation but the protein product of this gene was annotated as XP_013611767.1 in the NCBI database. This gene was identified as a member of the *BoHsf* gene family and had a highly conserved *Hsf* gene structure.

The physical and chemical properties of all 35 *BoHsf* genes were analysed (Table 1). The *BoHsf* proteins ranged from 227 amino acids (aa; *BoHsf31*) to 495 aa (*BoHsf05*) in length. The predicted isoelectric points (pI) varied from 4.66 (*BoHsf30*) to 9.27 (*BoHsf33*) and molecular weight (MW) from 26.14 kDa (*BoHsf27*) to 55.40 kDa (*BoHsf05*). The instability index, i.e. the stability of the protein in a test tube, indicated that all the *BoHsf* proteins are unstable except *BoHsf26* which had an instability index of 34.72 and was considered stable. *BoHsf* proteins were mostly composed of aliphatic amino acids, and *BoHsf23* and *BoHsf24* had the largest aliphatic index (78.5). The GRAVY values which reflect the hydrophobicity of the protein were all < 0 for all the *BoHsf* proteins, suggesting that all of them are hydrophilic. Some differences in the physical and chemical attributes of the *BoHsf* proteins can be observed. These differences are mostly due the difference in the non-conserved region amino acid composition.

Phylogenetic and evolutionary analysis of *B. oleracea* HSFs

The amino acid sequences of *Hsf* genes from *A. thaliana*, *B. rapa* and *B. napus* were used to construct a phylogenetic tree together with *B. oleracea* Hsfs. In this study, a total of 21 *Arabidopsis*, 36 *B. rapa*, 64 *B. napus* and 35 *B. oleracea* *Hsf* proteins were used. *Arabidopsis* was chosen because it is a model plant for dicots and it belongs to the Brassicaceae family (Price 1995). *Arabidopsis* shares recent common ancestry (20–24 MYA) with *Brassica* species (Ziolkowski et al. 2006). *B. rapa* and *B. napus* were chosen because *B. oleracea* and *B. rapa* have undergone hybridisation to form *B. napus*. One hundred and fifty-six *Hsf* proteins from four species were clearly divided into classes A, B and C, with well-supported bootstrap values (Fig. 1). The gene distribution within those classes was investigated. Most of the *BoHsf* genes belonged to

Table 1 Information of BoHsfs, including gene ID, type of Hsf, chromosomal location, protein sequence length, isoelectric point (pI), molecular weight (Mw), instability index (I.I), stability (U: unstable protein, S: stable protein) aliphatic index (A.I) and grand average of hydropathicity (GRAVY)

Name	Gene ID	Type	Chr. no.	No. of aa	pI	Mw	I.I	Stability	A.I	GRAVY
BoHsf01	Bo1g017810	A1a	C1:6265162–6,266,597	436	5.06	48,346.96	69.45	U	62.84	–0.613
BoHsf02	Bo9g161770	A1b	C9:47699173–47,701,363	466	4.76	51,659.37	53.99	U	70.24	–0.632
BoHsf03	Bo9g151630	A1b	C9:45050050–45,052,240	466	4.76	51,659.37	53.99	U	70.24	–0.632
BoHsf04	Bo00967s020	A1b	Unplaced Scaffold	466	4.76	51,659.37	53.99	U	70.24	–0.632
BoHsf05	Bo5g048950	A1d	C5:16251402–16,254,319	495	4.64	55,404.4	61.92	U	64.59	–0.783
BoHsf06	Bo5g150300	A1e	C5:46259612–46,261,838	448	4.91	49,626.37	51.02	U	76.92	–0.481
BoHsf07	Bo3g055950	A1e	C3:22158414–22,160,491	455	5.84	50,174.01	54.66	U	72.13	–0.544
BoHsf08	Bo1g155010	A1e	C1:43152430–43,154,774	453	5.04	50,195.78	55.05	U	69.8	–0.641
BoHsf09	Bo3g042080	A2	C3:17168238–17,169,672	349	5.09	39,249.14	48.76	U	71.75	–0.542
BoHsf10	Bo9g181780	A3	C9:53783895–53,785,735	401	5	45,024.58	51.04	U	64.41	–0.64
BoHsf11	Bo7g106320	A4a	C7:41208791–41,210,053	393	5.3	45,023.05	68.73	U	64.68	–0.886
BoHsf12	Bo3g157380	A4a	C3:56354756–56,355,934	362	5.13	41,733.58	64.73	U	64.59	–0.908
BoHsf13	Bo1g020010	A4c	C1:7225408–7,226,663	384	5.13	44,214.2	69.16	U	62.11	–0.912
BoHsf14	Bo9g062710	A4c	C9:18896025–18,897,122	340	5.64	39,053.6	53.76	U	69.06	–0.837
BoHsf15	Bo4g128450	A5	C4:33368868–33,370,417	475	5.67	52,690.37	60.29	U	64.11	–0.787
BoHsf16	Bo3g081420	A6b	C3:29769431–29,771,220	391	4.94	44,766.81	56.87	U	60.77	–0.864
BoHsf17	Bo5g093470	A6b	C5:30568737–30,570,796	405	4.94	46,587.64	58.1	U	58.94	–0.914
BoHsf18	Bo1g103760	A6c	C1:31682927–31,684,796	398	5.12	45,851.02	61.76	U	62.16	–0.937
BoHsf19	Bo7g101300	A7a	C7:39457917–39,459,453	270	5.76	31,139.43	56.91	U	78	–0.589
BoHsf20	Bo7g101330	A7a	C7:39472011–39,473,306	267	5.76	31,145.34	52.47	U	78.13	–0.63
BoHsf21	Bo4g125760	A7a	C4:32691062–32,692,689	268	5.75	31,153.26	61.37	U	69.14	–0.714
BoHsf22	Bo8g098580	A7b	C8:33792265–33,794,013	285	5.97	33,235.57	59.65	U	64.32	–0.727
BoHsf23	Bo6g106450	A8	C6:32979749–32,981,142	381	4.89	43,190.49	43.97	U	78.5	–0.678
BoHsf24	Bo2g056350	A8	C2:16701527–16,702,862	361	4.83	40,913.95	49.82	U	78.5	–0.628
BoHsf25	Bo1g005040	B1	C1:1137641–1,138,630	284	5.71	31,302.98	44.34	U	64.15	–0.642
BoHsf26	Bo7g118320	B1	C7:47425562–47,426,479	272	5.55	30,221.62	34.72	S	62.35	–0.724
BoHsf27	Bo2g161830	B2a	C2:51067305–51,068,091	233	6.35	26,148.26	70.75	U	65.24	–0.616
BoHsf28	Bo3g108110	B2a	C3:38631999–38,632,991	300	6.27	33,684.69	56.21	U	65.57	–0.648
BoHsf29	Bo3g045370	B2b	C3:19060542–19,061,584	319	4.9	34,730.4	57.48	U	65.42	–0.665
BoHsf30*		B2b	C9:28657306–28,659,316	361	4.66	38,237.14	53.36	U	66.54	–0.511
BoHsf31	Bo3g035750	B3	C3:13941016–13,943,276	227	5.88	26,600.81	48.32	U	64.76	–0.901
BoHsf32	Bo6g008100	B4	C6:1490252–1,491,338	300	8.42	37,657.5	49.63	U	67.94	–0.614
BoHsf33	Bo8g020430	B4	C8:4746425–4,747,532	335	9.27	38,322.22	48.78	U	64	–0.695
BoHsf34	Bo3g083520	C1	C3:30933687–30,934,634	315	6.06	35,791.15	47.74	U	62.79	–0.719
BoHsf35	Bo7g033000	C1	C7:12824213–12,825,241	315	5.71	36,124.52	47.33	U	65.52	–0.738

class A. The class A was further subdivided into nine sub-classes (A1, A2, A3, A4, A5, A6, A7, A8 and A9). Subclasses A1, A4, A6 and A7 were further divided into sub-sub classes. Eight *BoHsf* genes belonged to subclass A1, making it the largest subclass. An A9 subclass *Hsf* gene was found in *Arabidopsis* but not in *Brassica* species. *B. oleracea* did not have any *HsfA06a* genes. Class B *Hsfs* were subdivided into four subclasses (B1, B2, B3 and B4). Subclass B2 was further subdivided into subclasses B2a and B2b. Class C had only one subclass, i.e. C1 with no sub-subclass.

The phylogenetic analysis also showed the presence of new sub-subclasses in *Brassica* species. The subclasses A1, A4, A6, A7, B1, B2, B4 and C1 in *Brassic*as had more genes than in *Arabidopsis* and the new suggested sub-subclasses contained genes from all three *Brassica* species and no representative gene from *Arabidopsis* (Fig. 1). We have also noticed these sub-subclasses of

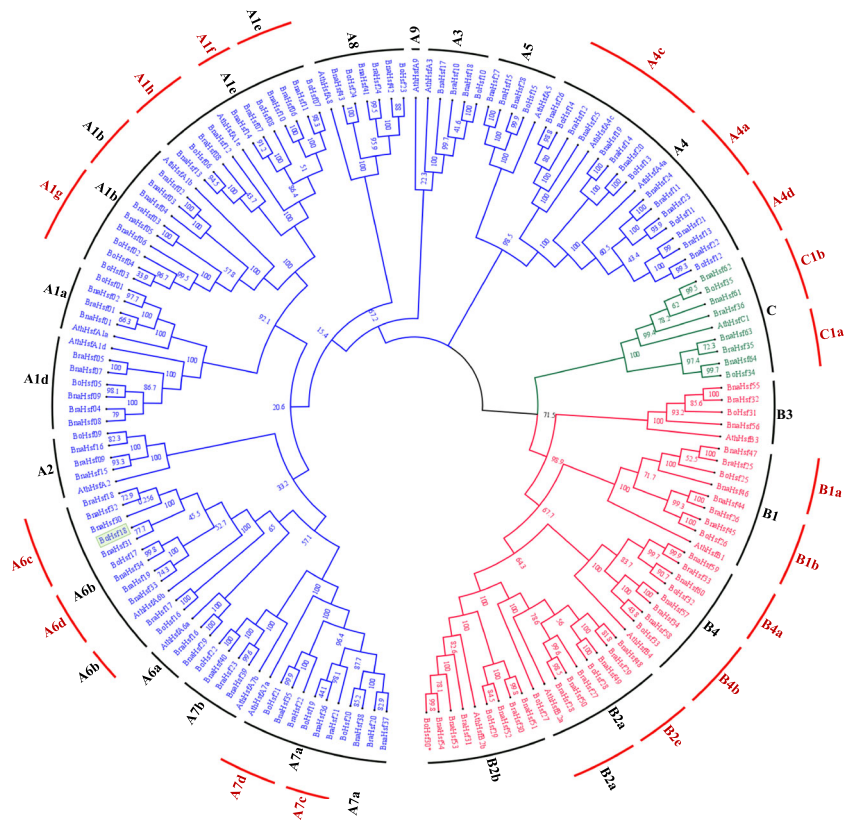
Hsf genes in *B. rapa* *Hsf* gene classification as downloaded from Heatster database (Scharf et al. 2012), which confirmed our results (Supplementary Table 2a–c).

The number of *Hsf* genes belonging to each class for *B. oleracea*, *B. rapa* and *B. napus* genomes were also compared (Fig. 2). The results suggest that some of the *Hsf* genes from *B. oleracea* and *B. rapa* are not present in *B. napus*. Although, the same number of the class C genes from *B. rapa* and *B. oleracea* are present in *B. napus*, this is not the case for the class A and class B *Hsf* genes.

Conserved domain and structural analysis of *B. oleracea* HSFs

We identified the conserved domains in *BoHsf* genes (Table 2). Heatster was also used for visualisation of the conserved domains in *BoHsf* genes (Fig. 3). The multiple alignments of the *BoHsf* protein sequences showed the

Fig. 1 The Neighbour-Joining phylogenetic tree of Hsf proteins from *B. oleracea*, *B. rapa*, *B. napus* and *Arabidopsis*. The phylogenetic tree was constructed by the MEGA 7.0 software using the neighbour-joining option with 1000 bootstrap replicates and pairwise deletion. Distinct colour segment represents different classes of Hsfs. Branches corresponding to partitions reproduced in less than 50% bootstrap replicates are collapsed. The evolutionary distances were computed using the Poisson correction method (Kumar et al. 2016) and are in the units of the number of amino acid substitutions per site. The analysis involved 156 amino acid sequences. All ambiguous positions were removed for each sequence pair. There were a total of 848 positions in the final dataset. The new proposed sub-classes are indicated by red text colour, and the type of *Arabidopsis* Hsf gene is mentioned in the gene name itself. Note: BoHsf18 is highlighted because it has been classified as Hsf6c



presence of the highly conserved DBD domain in all *BoHsf* genes (Fig. 4). The DBD domain is about 100 amino acids long and is located in the N-terminal of the *BoHsf* genes. The structure of the OD domain (HR-A/B region), which is also a conserved domain, serves as a basis for classification of Hsf proteins. Class B *BoHsf* proteins lack any insertion between the HR-A and HR-B regions (Fig. 5), while an insertion of 21 aa was found in class A and an insertion of 7 aa was found in class C *BoHsfs*.

The other domains found in Hsf proteins are the NLS, NES and AHA domains. NLS and NES are crucial for intercellular distribution and interactions of *Hsf* genes in the nucleus and the cytoplasm. NLS was predicted using the cNLS Mapper software and NES prediction was obtained by NetNES. Most of the *BoHsfs* contained NLS whereas, NES was not found in class C genes and some of the class B genes. AHA motifs were specific to majority of class A genes. Some of the class A proteins had two AHA motifs (*BoHsf09*, *BoHsf11*, *BoHsf12* and *BoHsf14*). To further refine the search for all the Hsf conserved domains, the Heatster database (Scharf et al. 2012) was used. The domains identified using Heatster were in agreement with our previous results from SMART, cNLS and NetNES

tools. Heatster also helped us identify the additional repressor domain (RD) in *BoHsfs*, which is specific to majority of class B proteins. Class B Hsfs are known to inhibit promoter activity through an active mechanism of repression that involves the C-terminal regulatory region (CTR). The RD domain is located in this CTR region. The presence of the RD domain in *BoHsf* class B proteins is in line with the repressor activity of these Hsfs.

Further, the structural diversity between *BoHsf* genes was compared in terms of exon/intron arrangement of the coding sequences. The number of introns of *BoHsf* genes ranged from zero to two. The detailed gene structure of the *BoHsf* genes is given in Fig. 6. Two introns were found in *BoHsf05* and *BoHsf22*, whereas zero introns were found in *BoHsf34*. All other *BoHsf* genes had only one intron. Most closely related *BoHsf* genes in the same class or subfamily shared a similar gene structure in terms of intron number and intron and exon length.

Expression profiles of *B. oleracea* Hsfs in different tissues

In order to understand the potential functions of *BoHsfs*, the tissue-specific expression of *BoHsfs* under non-stressed

Table 2 Functional domain analysis of BoHsf proteins

Name	Hsf type	DBD	HR-A/ B	NLS	NES	AHA1	AHA2	RD
BoHsf01	Ala	36–129	156–220	239–249 (MET:TEASICKRR)				
BoHsf02	Alb	25–118	141–206	227–237 (QIQGSSNKKRR)	439–446 (ITEQMIGTS)		390–399 (DPFWEQFFAD)	
BoHsf03	Alb	25–118	141–206	227–237 (QIQGSSNKKRR)	439–446 (ITEQMIGTS)		390–399 (DPFWEQFFAD)	
BoHsf04	Alb	25–118	141–206	227–237 (QIQGSSNKKRR)	439–446 (ITEQMIGTS)		390–399 (DPFWEQFFAD)	
BoHsf05	Al d	41–134	158–222	243–253 (RRISDTSKRR)	476–483 (LTQMGILLS)		382–391 (DPLCEQLFGE)	
BoHsf06	Ale	15–108	129–193	212–222 (QHVSEGNICKRR)	434–441 (LTEQMIGFLT)		389–398 (DSLWEQFFGE)	
BoHsf07	Ale	24–117	138–202	220–230 (QHISEGNICKRR)	441–448 (LTEQMIGFLT)		387–396 (DSLWEQFFGE)	
BoHsf08	Ale	22–115	136–200	219–229 (QHISEGNICKRR)	439–446 (LTEQMIGFLT)		328–330 (LDW)	
BoHsf09	A2	42–135	155–219	235–248 (ICKGLFGSDVGRICRR)	341–345 (QMGFLG)	277–285 (TLAAAIDD)		
BoHsf10	A3	49–142	173–219	246–255 (ICARICKFIECHS)				
BoHsf11	A4a	14–107	132–189			256–265 (IAIWENLVSDS)	343–352 (DVFWQQLLTEN)	
BoHsf12	A4a	10–103	126–183	202–205 (RKRR)		233–242 (IAVWENLVSED)	312–321 (DVFWQQLLTEN)	
BoHsf13	A4c	13–106	131–188	207–210 (RKRR)			330–339 (DVFWQQLLTEN)	
BoHsf14	A4c	11–104	128–185	200–203 (RICRIC)		228–237 (LTFLENLVTE)	289–298 (DVFWQQLT'EN)	
BoHsf15	AS	24–117	137–194	205–220 (RKVEGLDVSAYNKKRR)	463–470 (MSRNTKNI)		423–432 (DVFWQQLTER)	
BoHsf16	A6b	58–151	175–239	254–268 (ICKDMEEAIRICKRQR)	376–386 (VDVLIEQLGYL)		354–363 (EDFLEGLFKED)	
BoHsf17	A6b	58–151	175–239	254–268 (ICKEMEEAISICKRQR)	390–400 (VDVLIEQLGYL)		368–377 (EGFWEDMLNE)	
BoHsf18	A6c	55–148	172–236	251–265 (KSKDTEEAIDKKRQR)	383–393 (VDVLIEQLGYL)		360–369 (EGFWEDMLNE)	
BoHsf19	A7a	27–120	136–186		219–229 (VSELEALALEM)		251–259 (DDGFWEELL)	
BoHsf20	A7a	27–120	133–183		216–226 (VSELEALALEM)		248–256 (DDGFWEELL)	
BoHsf21	A7a	27–120	135–185		218–228 (MSELEALALEM)		250–258 (DDGFWEELL)	
BoHsf22	Alb	24–117	152–202		235–245 (VSELEALALEM)		261–269 (DDGFWEELL)	
BoHsf23	AS	18–126	155–212	304–312 (ICKKEMICVQE)			290–299 (DDGAWEKLL)	237–243 (KLFGVWV)
BoHsf24	AS	16–124	151–208	300–308 (ICKKNVICKGC)			286–295 (DDGAWEICLL)	224–230 (laFGVWV)
BoHsf25	B1	9–102	149–186	247–250 (RKRR)				
BoHsf26	B1	9–102	137–174	233–236 (RKRR)				
BoHsf27	B2a	22–115	165–201					
BoHsf28	B2a	21–114	172–208	260–264 (ICRIRS)				251–257 (RLFGVSI)
BoHsf29	B2b	34–127	188–224	269–273 (KRARR)				260–266 (RLFGVSI)
BoHsf30*	B2b	46–139	217–253	307–311 (KRARR)				298–304 (RLFGVSI)
BoHsf31	B3	38–131	172–207					220–225 (KLFGVK)
BoHsf32	B4	32–125	195–231	298–301 (KKRS)	313–325 (LVLDKSDGLNLNM)			288–294 (ICLFGVSL)
BoHsf33	B4	32–125	195–231	303–306 (KKRS)	318–330 (LVLDNSDLGLNLNM)			293–299 (ICLFGVSL)
BoHsf34	CI	17–111	123–166	188–193 (DKKRR)				
BoHsf35	CI	14–108	124–167	188–193 (DQICICRR)				

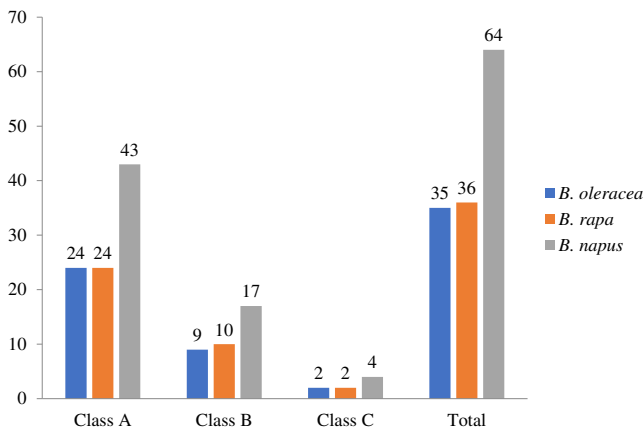


Fig. 2 Comparative analysis of the number of *Hsf* genes in *B. oleracea*, *B. rapa* and *B. napus*

conditions was examined. The *BoHsf* expression analysis was performed across seven different tissues (root, leaf, stem, flower, siliques, callus and bud) using RNA-Seq data (Supplementary Table 3). The expression heat map was used to visualise the *BoHsf*s tissue-specific expression patterns (Fig. 7). The TPM expression values of the *BoHsf*s were compared to study the level of gene expression in different tissues. The results suggest that *BoHsf*s have different expression patterns across tissues. All but one of the *BoHsf*s (*BoHsf19*) were expressed in at least one tissue analysed (TPM values of > 0 was used to consider a gene expressed in a given tissue), 8 *BoHsf* genes (*BoHsf02*, *BoHsf03*, *BoHsf04*, *BoHsf06*, *BoHsf13*, *BoHsf18*, *BoHsf19* and *BoHsf31*) were not expressed in flowers and two out of these, *BoHsf04* and *BoHsf31*, were not expressed in the reproductive tissues analysed. Moreover, the *BoHsf* genes showed a higher mean expression in vegetative tissue (root, stem and callus) than in reproductive tissues (bud and flower). Class C *BoHsf* genes were expressed in all tissues with a higher mean expression level in root, stem and callus.

Identification of orthologous gene clusters

Comparative analysis was performed to identify orthologous *Hsf* gene clusters across the four species, *A. thaliana*, *B. oleracea*, *B. rapa* and *B. napus*. This analysis was performed to evaluate the effect of polyploidisation events, which affected the evolution of the Brassica genomes, on the *Hsf* gene family. The orthologous gene clusters identified and their overlaps for the four species are illustrated in Fig. 8. The orthologous gene clusters were also identified between the Brassica species alone (Supplementary Fig. 1a) and between each of the Brassica species and *A. thaliana* separately (Supplementary Fig. 1b). The results indicate the presence of orthologous gene clusters, which were solely

composed of genes found in Brassica species, but not *A. thaliana*, suggesting that polyploidisation has resulted in formation of new Brassica-specific orthologous gene clusters. The results also indicate that with increasing phylogenetic distance, the number of identified orthologous genes decreases. Supplementary Table 4 provides a detailed list of orthologous gene clusters and singletons from the above analyses (Supplementary Table 4a–e).

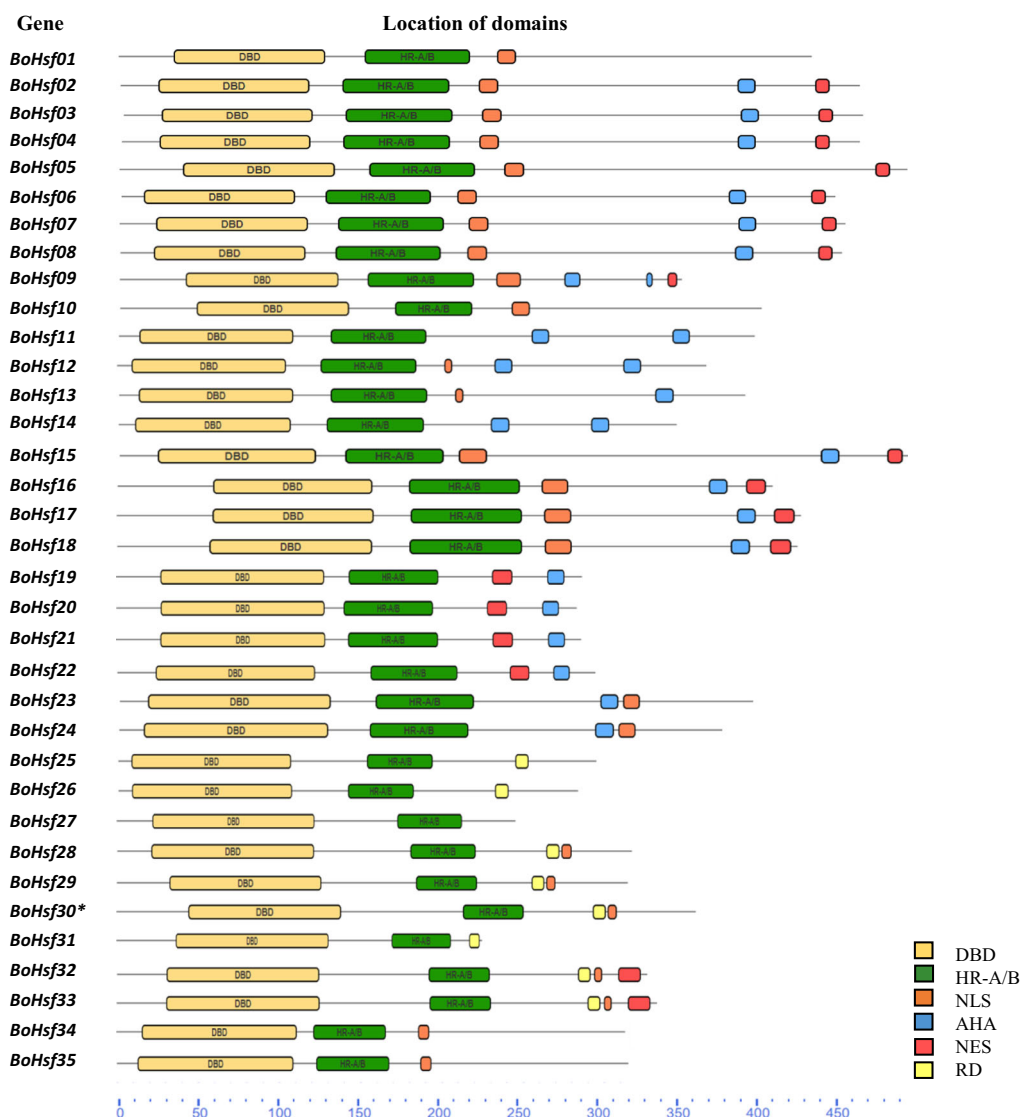
Gene duplication analysis

Gene family expansion can occur as a result of small-scale and global duplication events. Segmental duplications encompass individual genes or genomic regions and can be either tandem (the new gene/sequence found adjacent to the duplicated genomic region) or interspersed (the new gene/sequence found at some more distal inter- or intra-chromosomal location). The global events include copying of the entire genome. The sequence similarity and physical position along the chromosome along with syntenic information regarding relationships between the chromosomal segments can help pinpoint the origin of gene family members.

In *B. napus*, A_n and C_n genomes contained 32 *Hsf* genes each. The chromosomal location of 54, 34 and 34 *Hsf* genes were known from *B. napus*, *B. oleracea* and *B. rapa*, respectively. The syntenic relationships among the three species (*B. oleracea*, *B. rapa* and *B. napus*) are illustrated in Fig. 9. Among the *Hsf* genes, there were 26 syntenic genes on the A_n genome with *B. rapa* and 23 syntenic genes on the C_n genome with *B. oleracea*. The detailed summary of the comparison of protein identity of the syntenic genes is given in Supplementary Table 5a. We found that the syntenic genes on the C_n genome retained more than 90% protein identity except for *BnaHsf58*. And in the case of the A_n genome, almost all syntenic genes were > 90% identical at the protein level to the corresponding gene in *B. rapa*, except for *BnaHsf36*, *BnaHsf25* and *BnaHsf59*. The results indicate high conservation of the gene structure following hybridisation and polyploidisation.

In *B. oleracea*, we identified nine groups of duplicated genes out of which five groups had three genes and four groups had two genes (Supplementary Table 5b). The adjacent genes were analysed to determine whether tandem duplication had taken place. Tandem duplication might have resulted when a pair of genes is separated by few genes in a 100-kb region of a chromosome (Lin et al. 2014). From the *BoHsf19*, *BoHsf20* and *BoHsf21* triplet, *BoHsf19* and *BoHsf20* are found on the same chromosome and the pair was most likely formed due to tandem duplication, while *BoHsf21* was found in syntenic position on another chromosome. Eight of the groups

Fig. 3 Conserved domain locations on BoHsf proteins. The domains in BoHsf proteins were analysed and visualised using the Heatster database



contained genes which were syntenic (a result of whole genome duplication) and one pair (*BoHsf03* and *BoHsf02*) in a triplet group (*BoHsf02*, *BoHsf03* and *BoHsf04*) contained non-syntenic genes. The location of *BoHsf04* was not known so we cannot comment on the type of duplication this gene was involved in.

In *B. rapa*, we identified eight groups of duplicated genes out of which two groups had three genes and six groups had two genes (Supplementary Table 5c). *BraHsf20* and *BraHsf21* gene pair and *BraHsf27* and *BraHsf28* gene pair showed tandem duplication. Seven of the duplicated gene groups contained genes which were syntenic and one duplicated gene pair (*BraHsf34* and *BraHsf33*) contained non-syntenic genes.

In *B. napus*, we first identified the gene duplications within the A_n genome and the C_n genome. There were

six duplicated pairs of gene on the A_n genome. The physical location of one or both genes was not known in three duplicated gene pairs (*BnaHsf21*, *BnaHsf24*; *BnaHsf10*, *BnaHsf14*; *BnaHsf44*, *BnaHsf47*). Out of the remaining three duplicated gene pairs, two pairs were syntenic (*BnaHsf57*, *BnaHsf59*; *BnaHsf61*, *BnaHsf63*). The genes in the third pair, i.e. *BnaHsf36* and *BnaHsf37*, were found on the same chromosome, and the pair was most likely formed due to tandem duplication and was non-syntenic. Similarly, eight duplicated gene groups were identified on the C_n genome. One duplicated gene group had three genes (*BnaHsf20*, *BnaHsf22*, *BnaHsf23*) and rest of the groups had two genes. The physical location of one or both genes was not known in three of the duplicated gene pairs (*BnaHsf30*, *BnaHsf31*; *BnaHsf30*, *BnaHsf34*; *BnaHsf45*,

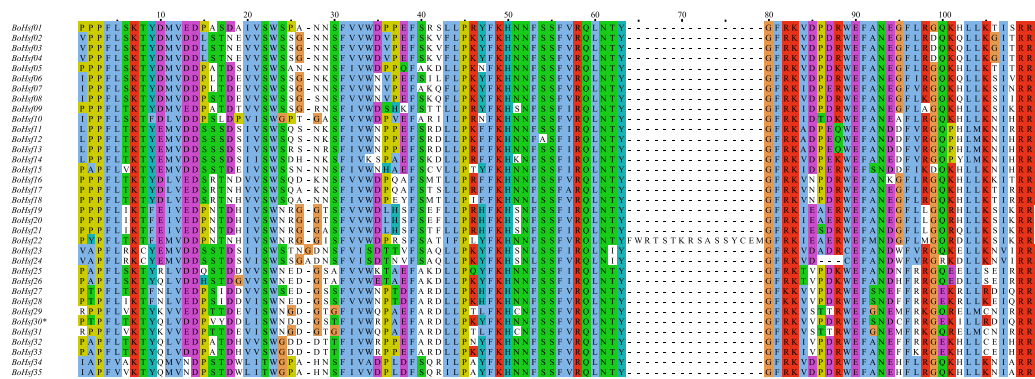


Fig. 4 Intron-exon structure of *BoHsf* genes. The intron-exon structures were produced by the GSDS2.0 online tool

BnaHsf46). Out of the remaining five groups, three duplicated gene groups (*BnaHsf20*, *BnaHsf22*, *BnaHsf23*; *BnaHsf35*, *BnaHsf38*; *BnaHsf42*, *BnaHsf43*) contained genes which were syntenic and two of the groups (*BnaHsf58*, *BnaHsf60*; *BnaHsf62*, *BnaHsf64*) contained genes which were non-syntenic.

Secondly, we identified gene duplications between the A_n and C_n genes. Forty-three duplicated gene pairs were identified. The physical location of one or both of the genes was not known in 12 duplicated gene pairs. The identified duplicated gene pairs showed the

presence of some pairs that shared common genes and contained syntenic genes. This was due to the fact that the pairs had genes that shared duplications within the A_n genome or C_n genome. Three pairs contained non-syntenic genes (*BnaHsf35*, *BnaHsf37*; *BnaHsf37*, *BnaHsf38*; *BnaHsf51*, *BnaHsf52*). Supplementary Table 5d provides a detailed account of gene duplications in *B. napus*.

In general, K_a (rate of nonsynonymous substitutions per nonsynonymous site) and K_s (rate of synonymous substitutions per synonymous site) values are commonly

Fig. 5 Multiple sequence alignment analysis of the DBD of *BoHsf* proteins. The DBD domain sequences were identified using SMART and Heatmap online tools and were aligned using the Clustal X2.0 software. The colours in the alignment are according to the ClustalX default colouring scheme

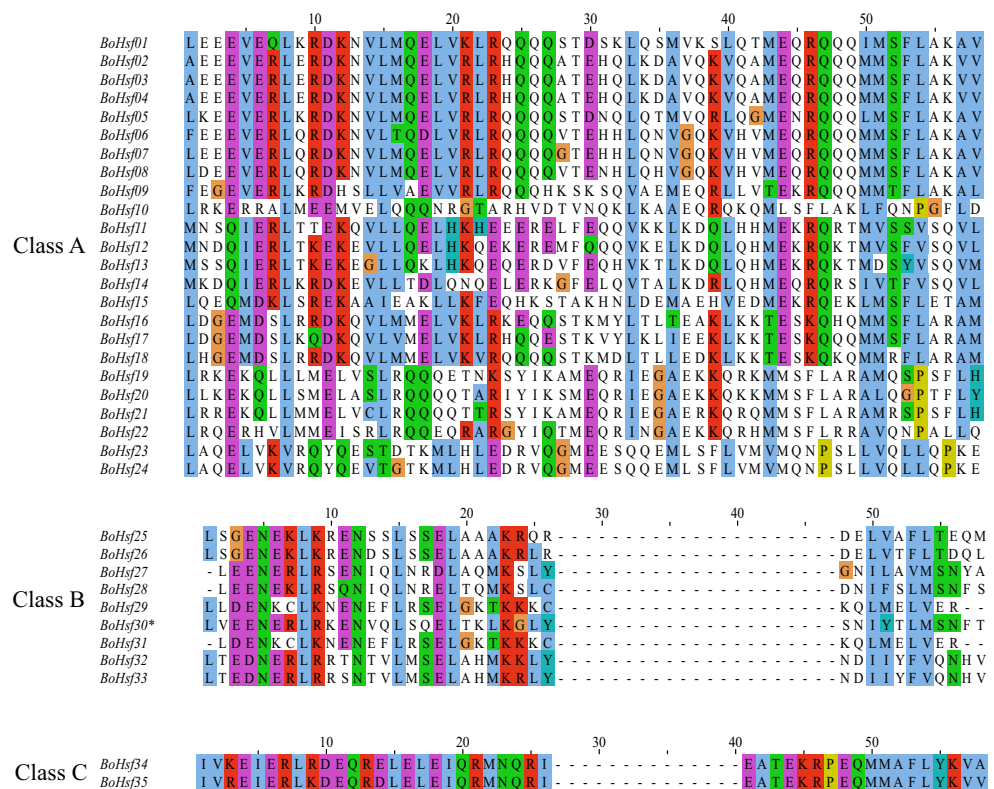




Fig. 6 Multiple sequence alignment analysis of the HR-A/B region (OD) of BoHsf proteins. The HR-A/B region sequences were identified using SMART and Heatster online tools and were aligned using the Clustal

X2.0 software. The colours in the alignment are according to the ClustalX default colouring scheme

used for assessing the rate of evolution in protein coding genes (Hurst 2002). The value of the Ka/Ks ratio represents the type of selection pressure on the gene and evolutionary rate. If Ka/Ks ~ 0 , then the selection is neutral, Ka/Ks < 1 indicates purifying selection and Ka/Ks > 1 indicates positive selection (Koonin and Rogozin 2003). The Ka/Ks ratios of the duplicated pairs of Hsfs in *B. oleracea*, *B. rapa* and *B. napus* were < 1 (refer to Supplementary Table 5b–d). The ratios suggest that the *Hsf* gene family in these species has evolved under purifying selection. Purifying selection is the removal of deleterious alleles and this type of selection helps in maintaining long-term stability of genes over the course of evolution (Koonin and Rogozin 2003). This suggests that the evolution of the *Hsf* gene family in the *Brassica* species (*B. napus*, *B. oleracea*, *B. rapa*) was governed by

strong constraints that may have contributed to their structural and functional stability.

Analysis of expression patterns of corresponding A and C genomes in *B. rapa*, *B. oleracea* and *B. napus*

The comparison of tissue-specific expression of *Hsf* syntenic genes was performed between the *B. napus* A_n genome and *B. rapa* and the *B. napus* C_n genome and *B. oleracea* (Fig. 10a, b). For the analysis of expression data, the Pearson correlation above 0.5 was considered as positive correlation and below -0.5 was considered as negative correlation (Supplementary Table 6a, 6b). The comparison of correlation coefficient values of tissue-specific expression of *Hsf* genes in leaf, stem, root and flower of *B. napus*, *B. oleracea* and

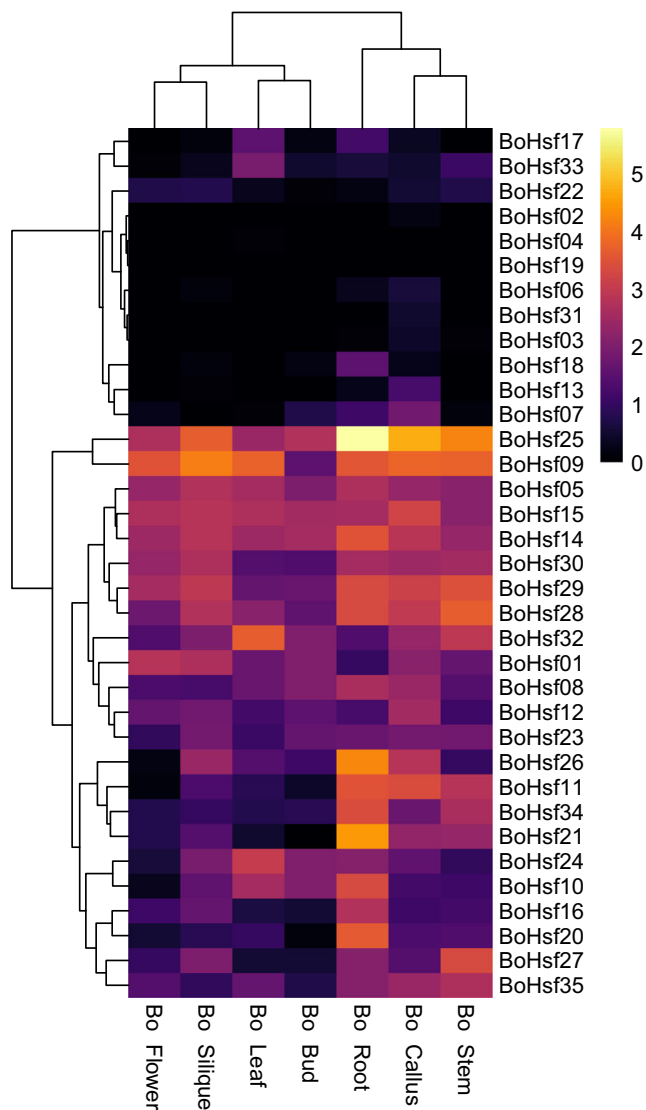


Fig. 7 Heat map representation and hierarchical clustering of *BoHsf* genes in root, stem, leaf, flower, silique, bud and callus. Colours correspond to $\log_1 p(\text{TPM})$

B. rapa indicated that 11 out of 23 syntenic genes showed positively correlated expression patterns between *B. napus* (C_n genome) and *B. oleracea* and 18 out of 25 syntenic genes between *B. napus* (A_n genome) and *B. rapa*. Two pairs of syntenic genes had a negative correlation value for gene expression data between *B. napus* and *B. oleracea*, and one pair of syntenic genes between *B. napus* and *B. rapa*. These negative correlations may be due to the difference between sampling time and sampling conditions.

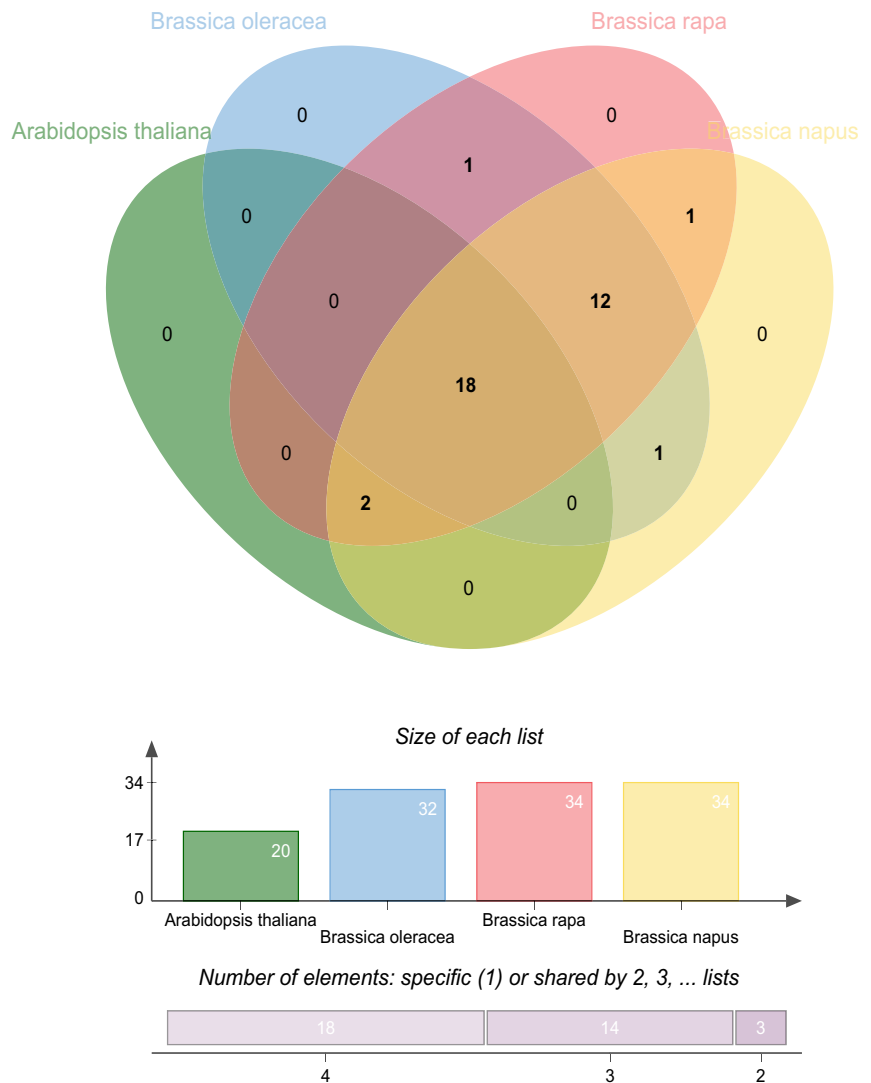
Discussion

B. oleracea is one of the important crops belonging to the genus *Brassica*. This *Brassica* species encompasses crops

like cabbage, cauliflower, broccoli, Brussels sprout, kohlrabi and kale (Golicz et al. 2016). The sequenced genome of *B. oleracea* (Liu et al. 2014; Parkin et al. 2014) provides a resource for the identification and characterisation of important gene families such as the heat shock factor (*Hsf*) gene family. The *Hsfs* acts as the terminal components of signalling networks involved in abiotic stress responses (Ohama et al. 2017; Saidi et al. 2011; Scharf et al. 2012). The *Hsfs* are the central regulators of HSPs expression involved in heat stress response (Baniwal et al. 2004; Jung et al. 2013). However, a study on the *Hsf* gene family in *B. oleracea* is lacking. We identified 35 putative *Hsf* genes in *B. oleracea* and a comparative analysis of the *Hsf* gene families in *B. oleracea*, *B. rapa* and *B. napus* indicated the whole genome duplication and hybridization events affecting the size of the *Hsf* gene family in *B. napus*. The *Brassica* genus has diverged from *A. thaliana* approximately 14–20 MYO (Beilstein et al. 2006; Li et al. 2004), and the *Brassica* species have undergone a whole genome triplication event, followed by gene duplications, deletions and rearrangements (Lysak et al. 2005). *B. napus* is an amphidiploid formed as a result of a hybridisation of *B. rapa* (A^rA^r , $2n=20$) and *B. oleracea* (C^oC^o , $2n=18$). The comparative analysis of the *Hsf* genes highlighted that the gene structure and domain compositions of the *Hsf* genes have been relatively conserved during evolution of the *Brassica* species. Though overall gene structure of A, B, and C *Hsf* genes classes were similar, the different classes of *BoHsf* proteins (A, B and C) also showed the presence of the characteristic domains. The analysis of phylogenetic and orthologous gene relationships also highlighted the presence of new sub-subclasses of *Hsf* genes in the three *Brassica* species, suggesting neo-functionalisation that remains to be confirmed by experimental evidence.

Our study of gene duplication revealed that gene duplications have contributed to the expansion of *Hsf* genes in the *Brassica* species. We identified one tandem duplication in *B. oleracea*, two in *B. rapa* and one additional in *B. napus* suggesting that most of the gene duplication events were large scale (Messing et al. 2004). Three genes in some of the duplicated gene groups were observed among *B. rapa* and *B. oleracea* *Hsfs*. The presence of three genes in the duplicated gene group may be due to the triploidisation event shared by these two species. We also identified duplicated gene groups which had two genes each, suggesting a gene loss event that may have accompanied the triploidisation (Song et al. 2014). The gene duplication analysis and comparative analysis of the members of the *Hsf* gene family among the three *Brassica* species showed that the structure of the largest *Hsf* gene family in *B. napus* was shaped by hybridisation, gene

Fig. 8 Orthologous gene clustering analysis between *B. oleracea*, *B. rapa*, *B. napus* and *Arabidopsis*. The orthologous gene clusters were identified and visualised using the OrthoVenn online tool



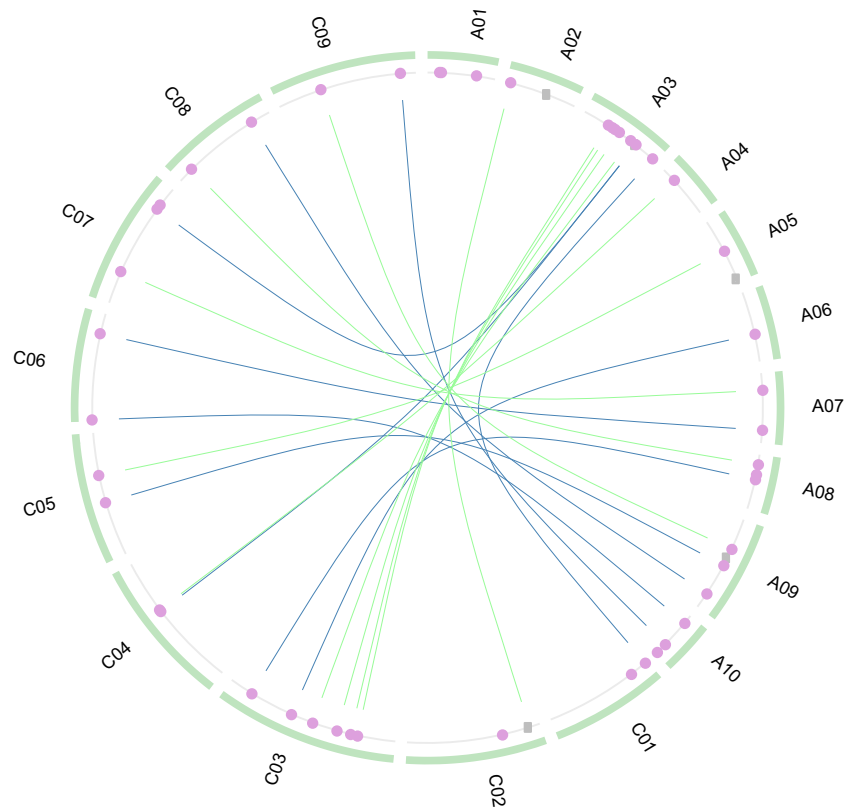
duplications and gene loss. The Ka/Ks values of the duplicated pair of genes suggested that they have undergone purifying selection. A similar purifying selection was observed by Lin et al. (2014) on *Glycine max* *Hsf* gene family. The purifying selection might have played a key role in maintaining the conserved structure of the *Hsf* genes throughout evolution.

Highly conserved syntenic genes were observed both structurally and regarding expression levels. Structural comparison of the syntenic genes highlighted several similarities between the *Hsf* genes in the three species. For instance, it has been reported in *B. napus* that the gene *BnaHsf63* and – 64 lacked an intron in the DBD which is characteristically present in all the other *Hsfs* (Zhu et al. 2017). The *BnaHsf63* gene is located on the A_n genome and is syntenic to *BraHsf35*. Similarly, the *BnaHsf64* gene is located on the C_n genome, and it is syntenic to the

BoHsf34 gene. Since *BraHsf35* and *BoHsf34* are the corresponding genes of *BnaHsf63* and *BnaHsf64*, respectively, they also lacked an intron in the DBD domain. Recently, Zhu et al. (2017) suggested that the difference in the structural organisation of the DBD domain may result in unique regulation and functional pattern for these genes.

The similarities between syntenic genes and expression level might suggest a high functional similarity. Previous studies have suggested the role of the *Hsf* genes of *B. rapa* and *B. napus* in heat, cold, drought, high levels of CO₂ and oxidative stresses (Zhu et al. 2017; Song et al. 2014). For example, *BnaHsf43* (A8) was shown to be upregulated in heat and high CO₂ treatment of *B. napus*. Therefore, it is attempting to speculate that the syntenic gene in *B. oleracea*, i.e. *BoHsf24*, might also be related to the response against

Fig. 9 Synteny analysis of *Hsf* genes in *B. oleracea*, *B. rapa* and *B. napus*. Purple dots: *Hsf* genes that have a match on the corresponding chromosome in *B. oleracea* or *B. rapa*; gray dots: *Hsf* genes that do not have a match on the corresponding chromosome in *B. oleracea* or *B. rapa*; green links: *Hsf* genes that have a match on the corresponding chromosomes; blue links: *Hsf* genes that have a match on different chromosomes



heat and high CO₂ stress. Similarly, tissue-specific expression patterns can also help us predict functional similarities between the homologous genes. For instance, *HsfA2* has been reported to play a key role in anther/pollen development in tobacco and tomato based on higher transcript levels in male reproductive cells. Also, *HsfA2* and *HsfA5* have been reported to be directly involved in the activation of protection mechanisms in anthers during heat stress and thus play a key role in stress tolerance during anther/pollen development and other stages of plant reproduction in tomato and Arabidopsis (Fragkostefanakis et al. 2016; Giorno et al. 2009; REŇÁK et al. 2014; Tunc-Ozdemir et al. 2013). We observed that the mean expression levels *HsfA2*, *HsfB1* and *HsfA5* homologues in *B. oleracea* (*BoHsf09*, *BoHsf25* and *BoHsf15*) were the highest in the reproductive tissues (flower and siliques). This suggests that along with *HsfA2* and *HsfA5*, *HsfB1* might also play a role in reproductive development and stress response in *B. oleracea*. Further, tissue expression analysis also indicated that *BoHsfs* might play a role during root development since the level of expression of most of the *Hsfs* (mainly *BoHsf25*, *BoHsf21*, *BoHsf26*) is highest in the roots. Our understanding of the evolution, structural and functional divergence of the *Hsf* gene family would enhance our understanding of the

pathways involved in plant stress responses and stress tolerance.

Conclusion

A total of 35 putative candidate *Hsf* genes were identified and characterised in *B. oleracea*. Study of the chromosomal locations, gene structure, structural organisation of conserved domains and tissue-specific expression patterns of these candidate genes provides a detailed account of the *Hsf* gene family in *B. oleracea*. The detailed phylogeny analysis of *Hsf* genes between *A. thaliana*, *B. oleracea*, *B. rapa* and *B. napus* suggests a high correlation between the evolution and a structural organisation of the *Hsf* gene family in these species. Some of the *Hsfs* such as *HsfA1*, *HsfA2*, *HsfA3*, *HsfA4*, *HsfA6a*, *HsfA9*, *HsfB1* and *HsfB2* have been functionally characterised in Arabidopsis (Guo et al. 2016) and have been proved to be an integral component of the signal transduction pathway in response to different stresses. Therefore, the orthologous gene candidates in the *Brassica* species can be further validated to confirm their role in various stress response. Further, a comparative analysis of the *Hsf* gene family in *B. oleracea*, *B. rapa* and *B. napus* pointed to the retention of the *Hsf* gene family following hybridisation.

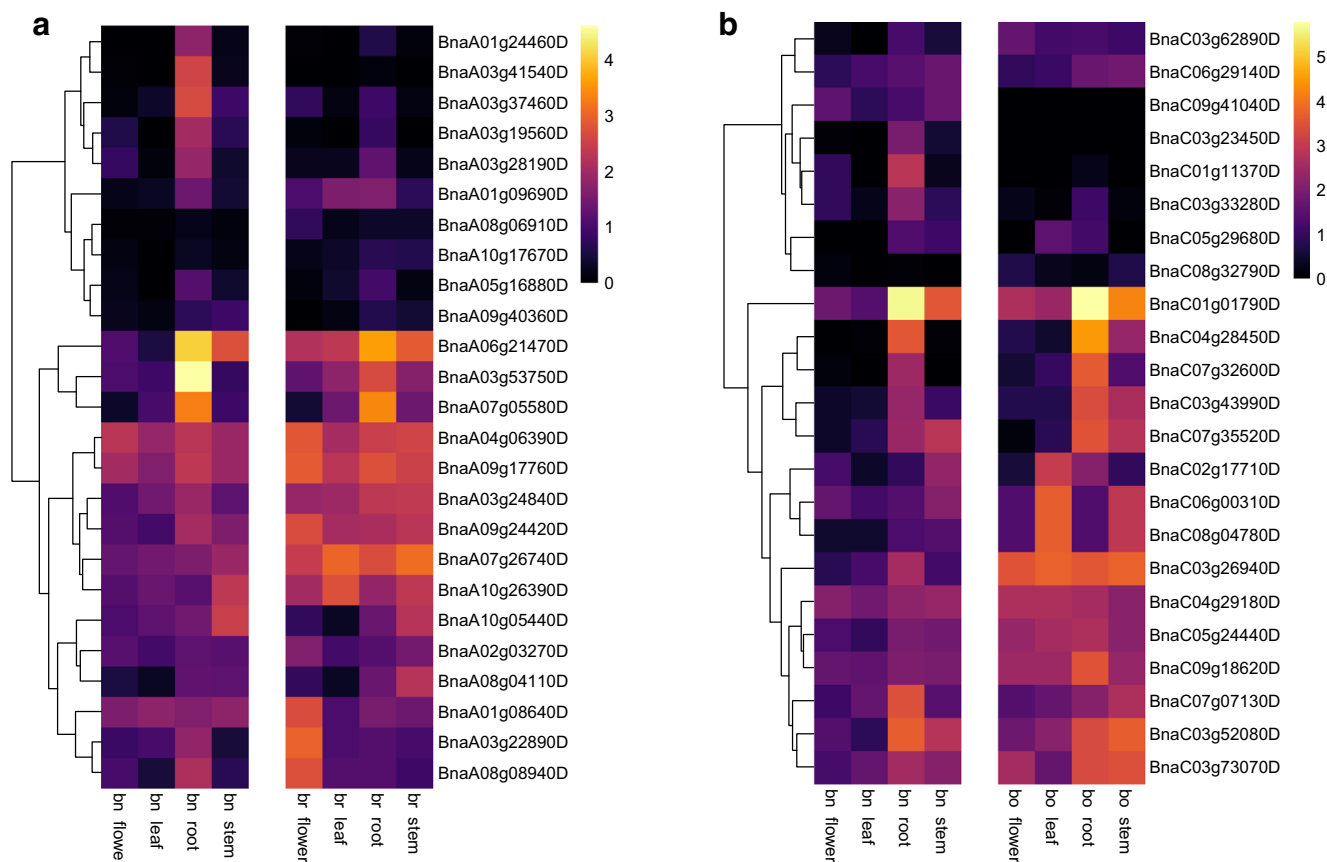


Fig. 10 **a** Heat map representation and hierarchical clustering of syntenic genes between *B. rapa* and *B. napus* in root, stem, leaf and flower. Colours correspond to $\log_1 p(\text{TPM})$. **b** Heat map representation and

hierarchical clustering of syntenic genes between *B. oleracea* and *B. napus* in root, stem, leaf and flower. Colours correspond to $\log_1 p(\text{TPM})$

The high conservation of syntenic order and physical location, as well as the similarity of structure of individual genes and expression profiles between the *B. napus* *Hsf* genes and the corresponding genes from *B. oleracea* and *B. rapa*, was observed. High-level synteny was also observed within *B. napus* *Hsf* genes between the A_n and C_n genomes. Gene duplication analysis of *Hsf* genes in *B. oleracea*, *B. rapa* and *B. napus* suggests that this gene family has undergone strong purifying selection thus resulted in conservation of the structure of the *Hsf* gene family. Together, the analyses suggest a strong functional conservation between the *Hsf* genes in the three *Brassica* species. In summary, this study provides a detailed account of *B. oleracea* *Hsf* genes and an understanding of the divergent retention, duplication and expansion of *Hsf* genes in the three *Brassica* species that resulted in the largest *Hsf* gene family in *B. napus*. This study also provides a future direction for validating the role of different *Hsf* genes in stress tolerance in the three agronomical important *Brassica* species. The identified genes can thus be focussed for developing crop varieties resilient to various environmental stresses.

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